MILITARY SPECIFICATION SHEET

ELECTRON TUBE, VIDICON

TYPE 7325

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of MIL-E-1.

DESCRIPTION: One inch, for film or live pickup with color or black-and-white TV cameras

PIN CONNECTIONS AND DIMENSIONS: See figure 1

ABSOLUTE-MAXIMUM RATINGS: See note 1

Parameter:	Ef	DIsj	IsjT	Ec3	Ec2	Ec1	Ehk	FI
Unit:	V	μAdc	μa	Vdc	Vdc	Vdc	Vdc	ft-c
Maximum:	6.9	0.25	0.55	750	750	-125	125	1,000
Minimum:	5.7	---	---	---	---	0	-10	---
TEST CONDITIONS:	6.3	(see note 2)	(see note 3)	(see note 4)	300	(see note 5)	---	(see note 3)

ABSOLUTE-MAXIMUM RATINGS: See note 1

Parameter:	Operating T(faceplate)	Storage temp.	tk	I(focus coil)	Alt
Unit:	°C	°C	sec	mAdc	ft
Maximum:	71	125	---	---	60,000
Minimum:	0	---	---	---	---
TEST CONDITIONS:	29 to 35	---	300	39	---

TEST PATTERN: EIA resolution chart, 1956, or equivalent (see note 6)

TEST RASTER: See note 7

TEST COMPONENTS: See note 8

REQUIREMENT OR TEST:

GENERAL

3.2 Qualification - Required

E-50.2 Holding period (t = 48 hours)

Section 5 - Preparation for delivery

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METHOD OR PARAGRAPH	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
						MIN	MAX	
	Qualification							
1331	Direct interelectrode capacitance	Target to all; no voltages	---	---	C	3.5	5.5	pf
1002	Barometric pressure, reduced	54 mm Hg	---	---	---	---	---	---
	Quality conformance inspection, part 1							
Appendix D, 20(b)	Dimensions	See figure 1	---	---	---	---	---	---
---	Alinement horizontal	Esj/DIsj = 0.02 μAdc (see notes 9, 10, and 11)	0.65	II	---	-36	+36	mAdc
---	Alinement vertical	Esj/DIsj = 0.02 μAdc (see notes 9, 10 and 11)	0.65	II	---	-36	+36	mAdc
---	Cutoff	Esj/DIsj = 0.02 μAdc; FI/IsjT = 0.22 μAdc (see notes 9, 11, and 12)	0.65	II	Ec1	-45	-100	Vdc
---	Center resolution (1)	Esj/DIsj = 0.02 μAdc; FI/IsjT = 0.22 μAdc (see notes 9, 11, and 13)	0.65	II	---	650	---	Lines
---	Corner resolution	Esj/DIsj = 0.02 μAdc; FI/IsjT = 0.22 μAdc (see notes 9, 11, and 14)	0.65	II	---	450	---	Lines
---	Spurious signal	Esj/DIsj = 0.02 μAdc; FI/IsjT = 0.22 μAdc (see notes 9, 10, 11, and 15)	0.65	II	---	---	---	---
---	Maximum dark current uniformity	Esj/DIsj = 0.2 μAdc; FI = 0 ft-c (see notes 9, 11, and 22)	0.65	II	ΔDIsj	---	23	%
---	Target volts (1)	Esj/DIsj = 0.02 μAdc; FI = 0 ft-c (see notes 2, 9, and 11)	0.65	II	Esj	15	---	Vdc
---	Sensitivity (1)	Esj/DIsj = 0.02 μAdc; FI = 1.0 ft-c (see notes 9, 11, and 16)	0.65	II	Isj	0.15	---	μAdc
---	Lag	Esj/DIsj = 0.02 μAdc; FI/IsjT = 0.22 μAdc (see notes 9, 11, and 17)	0.65	I	Lag	---	25	%
---	† Maximum grid No. 2 current	Esj = open; Ec1 = 0 Vdc; Ec3 = 250 Vdc	0.65	II	Ic2	1200	---	μAdc
---	† Gas ratio	See note 18	0.65	II	Gr	---	0.12	nAdc/ μAdc
---	Center resolution (2)	Esj/DIsj = 0.2 μAdc; FI/IsjT = 0.4 μAdc (see notes 9, 11, 13, and 21)	0.65	II	CeRes	500	---	Lines
---	After image	Esj/DIsj = 0.2 μAdc; FI/IsjT = 0.4 μAdc (see notes 9, 11, and 19)	0.65	II	AIm	---	3	sec

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METHOD OR PARAGRAPH	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
						MIN	MAX	
	Quality conformance inspection, part 1 -Continued							
---	Target volts (2)	Esj/DIsj = 0.1 μAdc; FI = 0 ft-c (see notes 2, 9, and 11)	0.65	II	Esj	---	90	Vdc
	Quality conformance inspection, part 2							
1301	Heater current		6.5	L6	If	540	660	mA
1105	Permanence of marking		---	---	---	---	---	---
Appendix D, 20(b)	Dimensions	See figure 1	---	---	---	---	---	---
	Quality conformance inspection, part 3							
4.7	Life-test provisions	Group D (see note 20) (t = 500 hours)(min)	---	---	---	---	---	---
4.7.3	Life-test end point: Maximum grid No. 2 current		---	---	Ic2	900	---	μAdc

NOTES:

1. The symbols used in this specification, in addition to those in MIL-E-1, are defined as follows:

   Lag = Percentage of initial value of signal-output current remaining after a specified time period following removal of illumination.
   Esj = Target voltage
   Isj = Signal-output current
   DIsj = Dark current
   IsjT = Target current (sum of Isj and DIsj)
   FI = Faceplate illumination
   T (faceplate) = Temperature of faceplate
   I (focus coil) = Focus coil current

2. With zero incident light on tube face, the target voltage (Esj) shall be adjusted to obtain the specified dark current (DIsj) (Esj/DIsj), as specified under test conditions.

3. The faceplate illumination (FI) shall be adjusted to obtain specified target current (IsjT) (FI/IsjT) as specified under test conditions. This may be done by adjustment of the lens opening. IsjT is defined as the sum of the signal-output current component (Isj) (0.20 μamp) and the dark-current component (DIsj).

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NOTES:-Continued

4. Grids No. 3 and No. 4 are connected together internally. The voltage applied to these two elements shall be referred to as Ec3. Ec3 voltages apply with the use of the specified focusing coil having a focus coil current of 39 mAdc, producing a field strength of approximately 40 gauss. Ec3 and optical focus shall alternately be adjusted to obtain best focus (sharpest lines).

5. Ec1 shall be adjusted until the beam current is just sufficient to discharge the photo-layer as observed by presence of detail in the highlights of the monitor.

6. The specified test pattern shall be placed on a transparent slide and shall be illuminated from the rear of the slide.

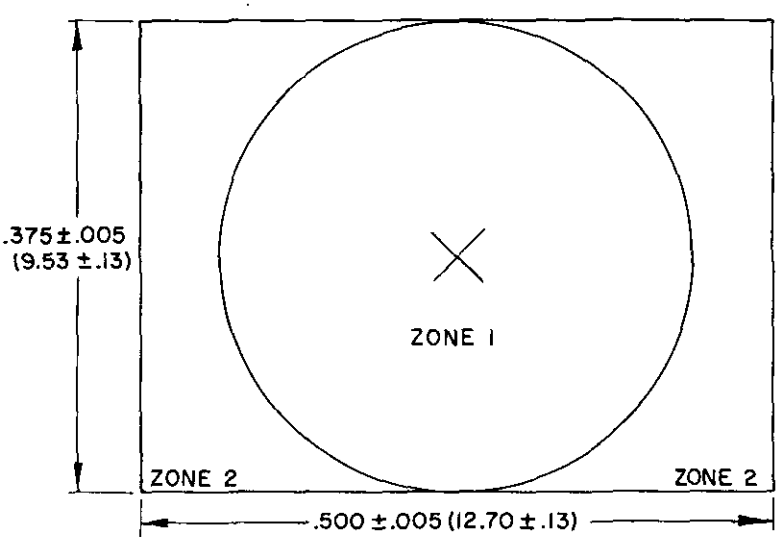
7. Standard EIA (RETMA) television scan shall be employed. The scan size shall be 1/2 x 3/8 inch.

8. Test components.

   (a) Deflection yoke - Cleveland Electronics No. VY-109-1AS, or equivalent.
   (b) Focusing coil - Cleveland Electronics No. VF-115-7, or equivalent.
   (c) Alinement coil - Cleveland Electronics No. VA-132, or equivalent.
   (d) The above components shall be mounted as shown on figure 3.

9. The tube shall be tested in camera with associated monitors as shown on the block diagram of figure 4

10. The following "back illuminated" test pattern shall be imaged on the face of the vidicon when performing this test:



NOTES:
a. All dimensions in inches.
b. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.

11. The alinement coil currents of each horizontal and vertical coil shall be adjusted until the center of the test pattern remains stationary as Ec3 is varied in and out of focus. The alinement coil currents shall be within the specified limits.

12. With test conditions as indicated, remove blanking voltage from grid No. 1 and adjust Ec1 until test pattern just disappears as observed on monitor.

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NOTES:-Continued

13. With Ec3 adjusted for balanced horizontal and vertical resolution, the resolution in the center of the pattern shall be not less than the specified limit.

14. With Ec3 adjusted for balanced horizontal and vertical resolution, the resolution in the four corners of the pattern shall be not less than the specified limit.

15. Spurious signal criteria.

(a) To be considered a black spot, the contrast ratio must be greater than 2.0 to 1.0.
(b) To be considered a white spot, the contrast ratio must be greater than 1.5 to 1.0.
(c) Spot size is measured by one of the three methods shown below.

METHOD I	METHOD II	METHOD III	NUMBER ALLOWED	
Measurement in inches using a monitor with 10-3/4 x 8-1/8 inch raster	Percent of raster height	Number of raster lines	Zone 1	Zone 2
Over 0.065	Over 0.80	Over 4	None	None
0.065 to but not including 0.050	0.080 to but not including 0.62	4 but not including 3	None	1
0.050 to but not including 0.015	0.62 to but not including 0.19	3 but not including 1	2	3
0.015 and under	0.19 and under	1 and under	*	*

*Do not count spots of this size unless concentration causes a smudge appearance.

(d) Maximum concentration of spots shall be evaluated by one of the three methods shown below:

Method I: Minimum separation between any two spots greater than 0.015 inch is limited to 1/4 inch.

Method II: Minimum separation between any two spots greater than 0.19 of raster height is limited to a distance equivalent to 3.1 percent of raster height.

Method III: Minimum separation between any 2 spots greater than 1 raster line is limited to 16 lines.

(e) Tubes are rejected for smudge, lines, streaks, mottled background, grainy background, or uneven background having contrast ratios greater than 1.5 to 1.0.

16. Project specified faceplate illumination on vidicon target. The lamp current shall be adjusted for a color temperature of 2,870° K. Sensitivity is the target current under these conditions minus dark current.

17. Lag test.

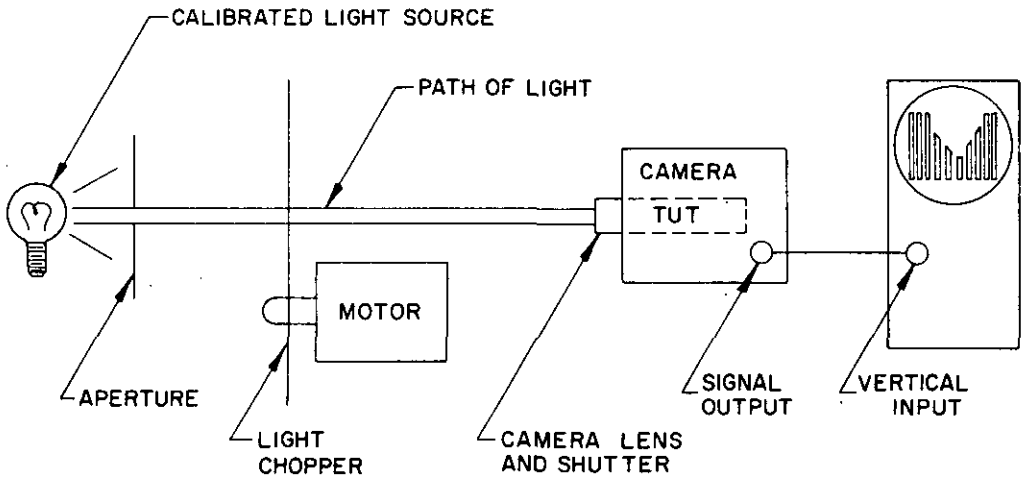
(a) Equipment
(1) Calibrated light source of 2,870° K color temperature.
(2) Suitable lens and apertures to obtain specified pattern and faceplate illumination.
(3) Motor driven light chopper capable of interrupting the path of light four times per second. The light chopper shall provide light to the vidicon target for 1/5 second and interrupt for 1/20 second such that the vidicon target receives light for about 12 TV fields and receives no light for about 3 TV fields.
(4) An oscilloscope capable of monitoring the signal output including the dc component. (P7 screen is recommended).

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NOTES:-Continued

17. Lag test. -Continued

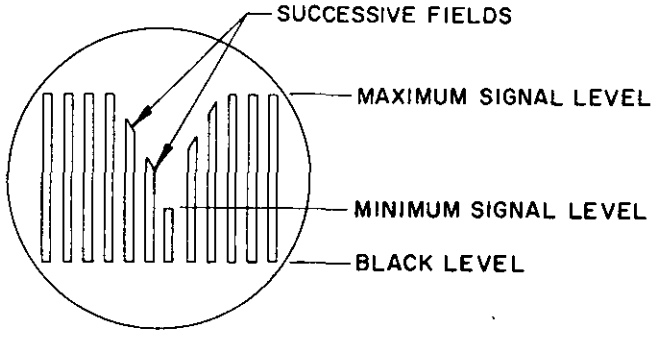
(b) Block diagram:



(c) Lag test procedure:

With conditions as indicated, and with test setup as shown on block diagram of note 17b, adjust oscilloscope sweep speed to obtain stationary presentation of approximately 12 fields similar to that pattern shown below. Remove incident light and set oscilloscope for "0" or no signal reference position. Turn "on" motor for light chopper, restore light level to tube, and adjust oscilloscope gain so that maximum signal level is 100 percent. (This is an arbitrary setting of the oscilloscope usually covering five divisions of the oscilloscope graticule.) The light chopper should now be adjusted so that the transition from light to dark occurs during vertical retrace.

OSCILLOSCOPE PRESENTATION



18. Gas ratio shall be determined as follows: With no light incident on the tube face, focus coil current off, Esj open, Ec1 = 0 Vdc, the following tests shall be made:

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NOTES:-Continued

18. -Continued

Ef	Ec3	Test procedure	Units
6.3 V	270 Vdc	Record G3 current as PIC3.	μAdc
6.3 V	-100 Vdc	Record G3 current as NIC3 when leakage becomes relatively stable but not necessarily zero.	nAdc
0 V	-100 Vdc	Turn filament off and record G3 current as ILC3 when G2 current drops to 0.5 mAdc.	
Gas ratio (Gr) =		(NIC3-ILC3)/PIC3	nAdc/μAdc

19. Using 0.2 μa I_{Dark}, apply signal for a period of 30 seconds using RETMA No. 1956, or equivalent, test pattern. Cap lens and note time required for signal to disappear.

20. Life-test conditions.

The vidicon shall be operated for the time specified under the following conditions:

Ef = 6.3 V
E_{G1} set to produce 0.5 ma I_{G2}
E_{G2} = +300 Vdc
Ambient temperature 40° C ± 5° C

21. Cap lens and adjust black level for 20 percent setup. Uncap lens and adjust video gain control to see all steps of grey scale on resolution chart.

22. The oscilloscope is adjusted looking at horizontal waveform, so that dark current signal above zero current (beam off) level is 100 percent on graduated scale. Total variation in dark signal is measured as a percentage of total signal.

Custodians:
Army - EL
Navy - SH
Air Force - 11

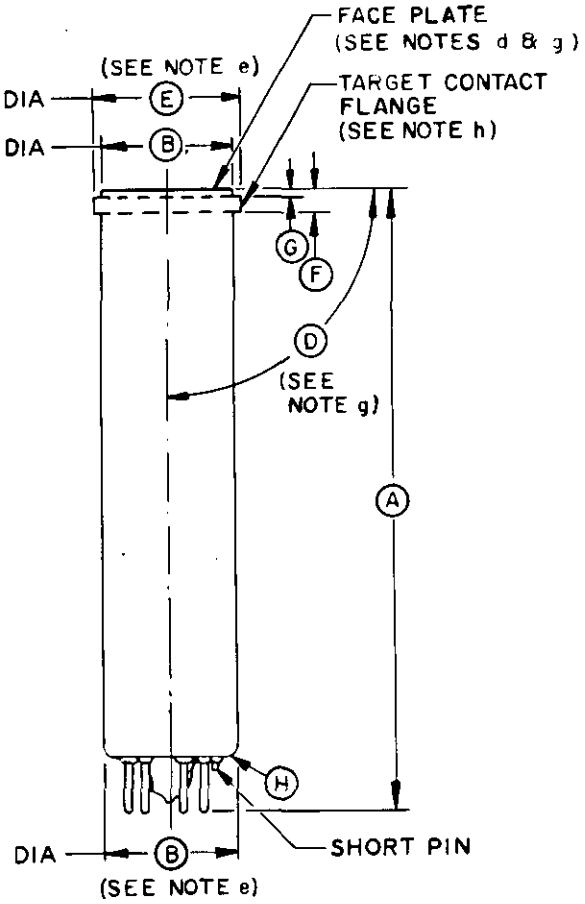
Preparing activity:
Navy - SH

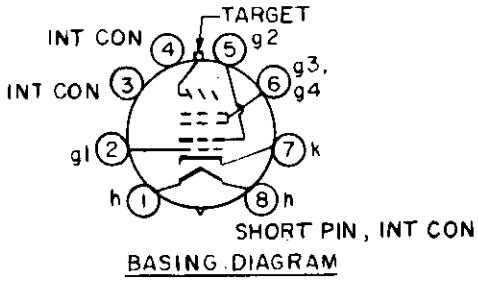
(Project 5960-2109)

Review activities:
Army - EL
Navy - SH
Air Force - 11, 17, 85

User activities:
Army - MU
Navy - WP, MC, CG
Air Force - None

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BASING DIAGRAM

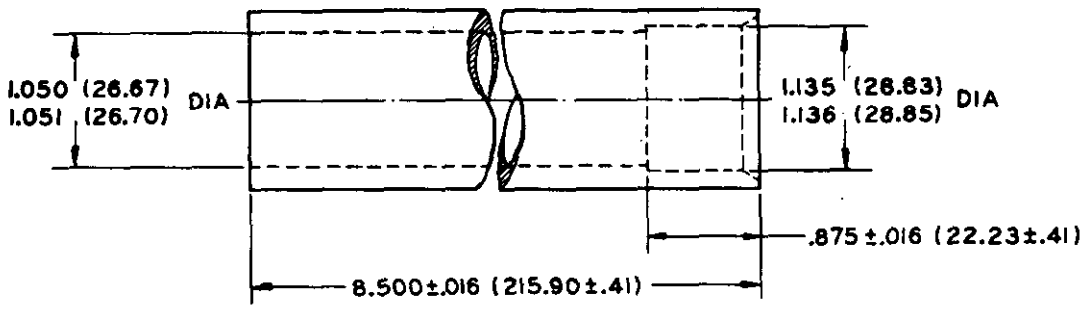
LTR	DIMENSIONS				NOTES
	INCHES		MILLIMETERS		
	MIN	MAX	MIN	MAX	
Qualification					
H	Base: E8-11				f
Quality conformance inspection, part 1					c
A	6.0	6.5	152.4	165.1	
B	.990	1.045	25.15	26.54	
E	1.116	1.132	28.34	28.75	
Quality conformance inspection, part 2					
D	89¼°	90¼°			
Nominal dimensions					b
F	.175		4.45		
G	.050		1.27		

NOTES:

a. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
b. These dimensions are for information only and are not required for inspection purposes.
c. The AQL for the combined mechanical defectives in quality conformance inspection, part 1 , shall be 1.0 percent, Inspection level I.
d. Direction of incident light is into face end of tube.
e. Concentricity and camber of tube shall be determined by gage shown on figure 2.
f. JEDEC designation. For pin alinement, use JEDEC gage GE8-2.
g. Faceplate tilt shall be gaged about entire circumference.
h. Target contact flange in form of metal ring, and with diameter as specified for dimension E, may be located along tube axis in any part of, or all of, space between dashed lines.

FIGURE 1. Outline drawing of electron tube type 7325.

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NOTES:

1. All dimensions in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. Millimeters are in parentheses.

FIGURE 2. Gage, concentricity, and camber.

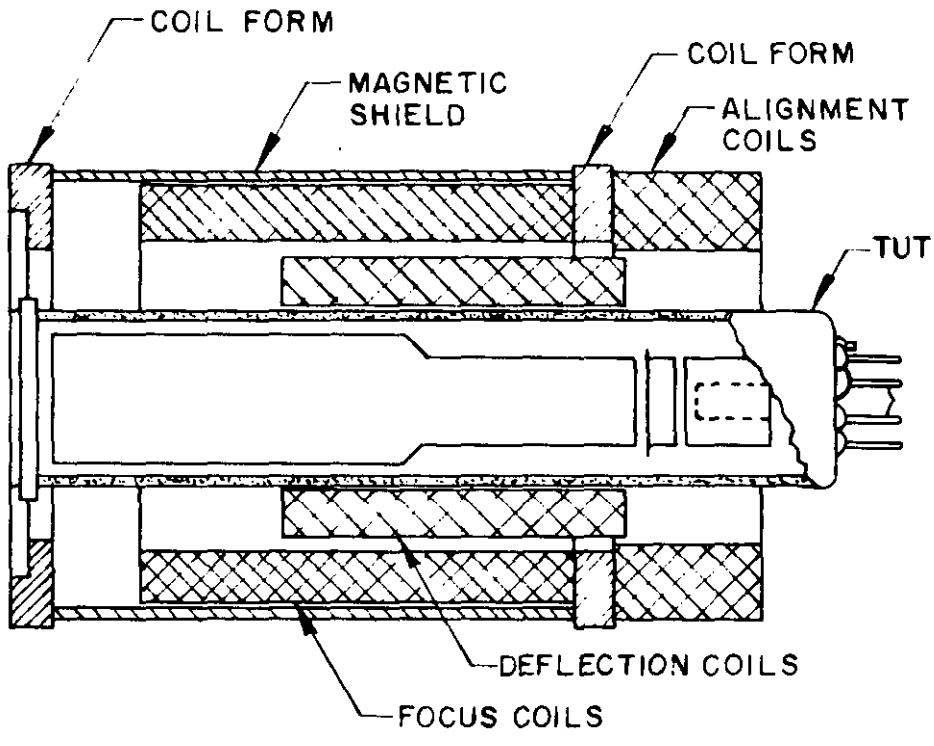
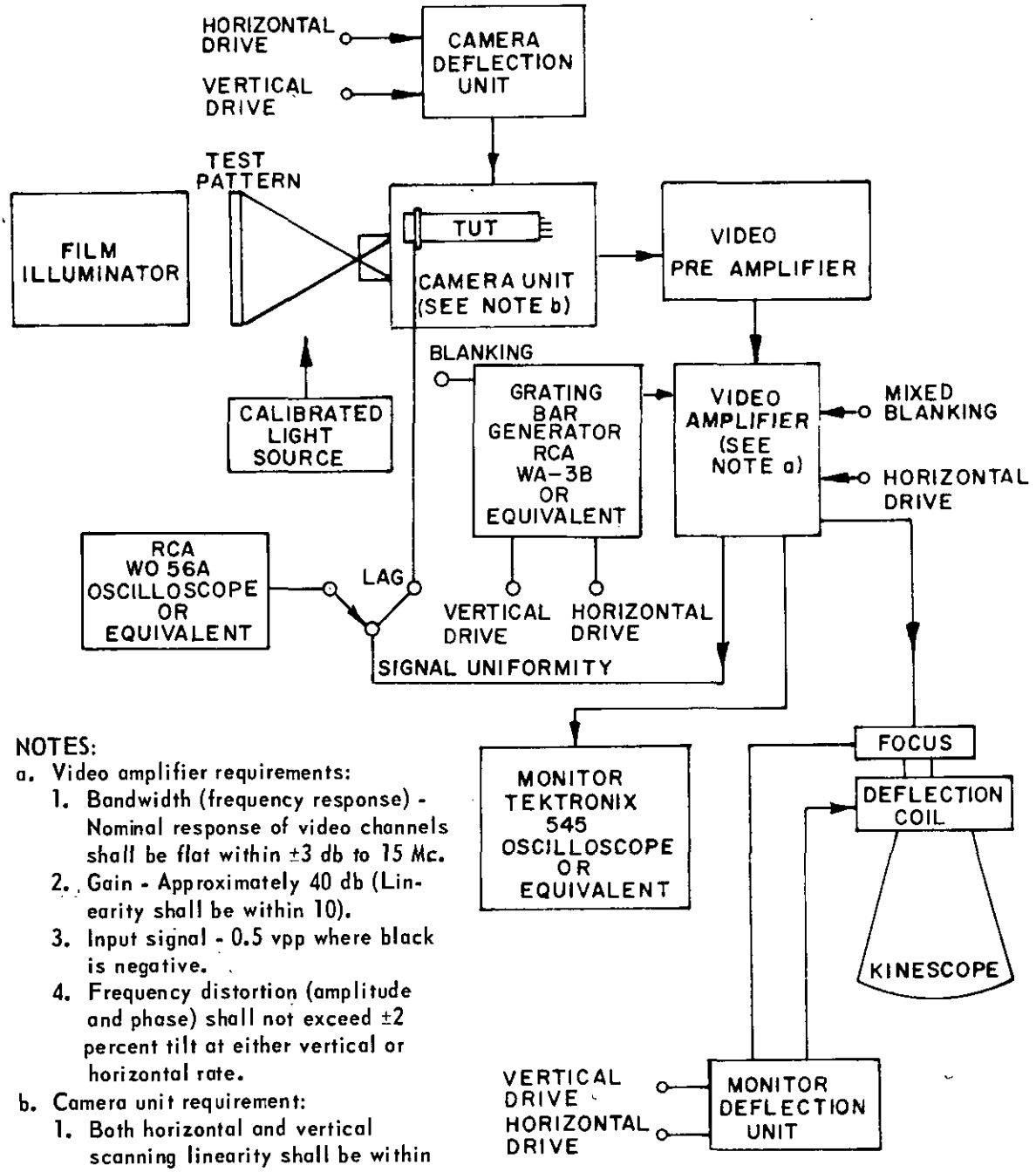


FIGURE 3. Test components.

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NOTES:

a. Video amplifier requirements:
   1. Bandwidth (frequency response) - Nominal response of video channels shall be flat within ±3 db to 15 Mc.
   2. Gain - Approximately 40 db (Linearity shall be within 10).
   3. Input signal - 0.5 vpp where black is negative.
   4. Frequency distortion (amplitude and phase) shall not exceed ±2 percent tilt at either vertical or horizontal rate.

b. Camera unit requirement:
   1. Both horizontal and vertical scanning linearity shall be within 1 percent.

FIGURE 4. Block diagram of test equipment.

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